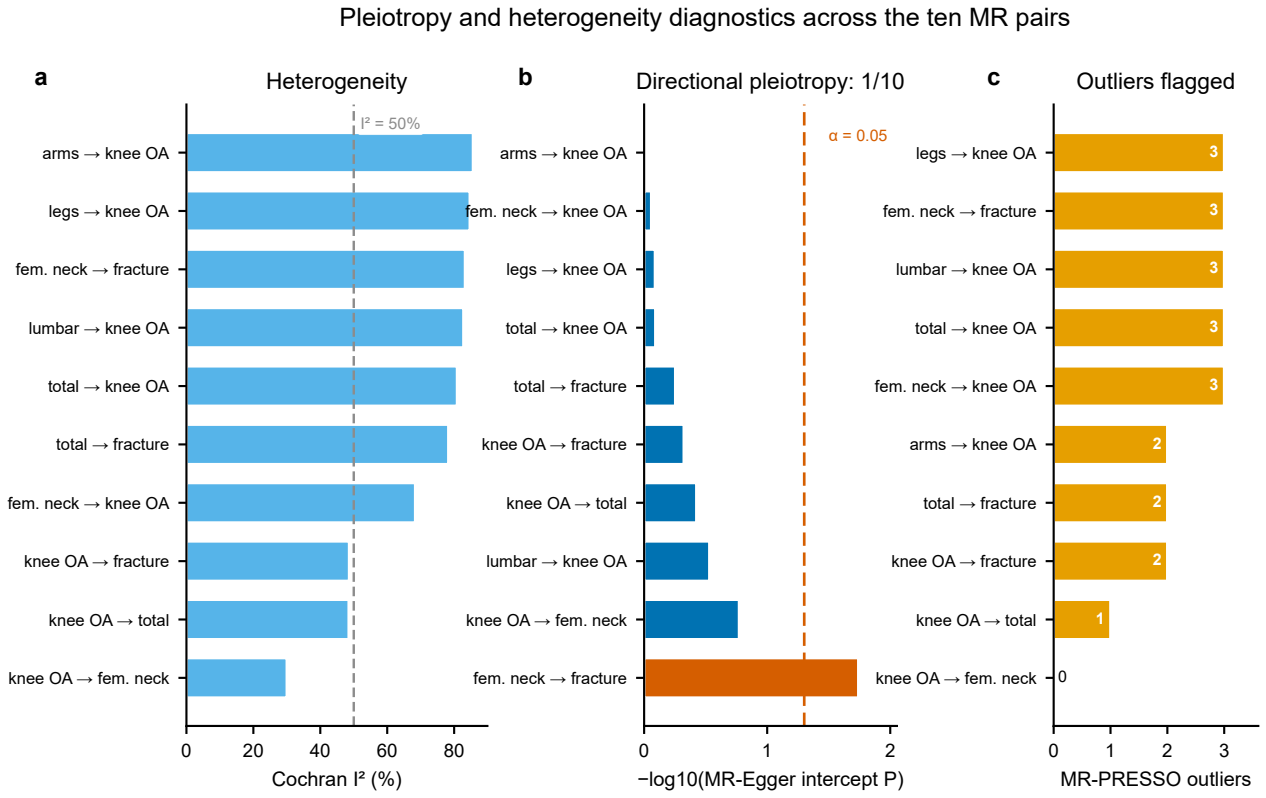
Pleiotropy and heterogeneity diagnostics across the ten MR pairs
a
Heterogeneity
I² = 50%
arms → knee OA
legs → knee OA
fem. neck → fracture
lumbar → knee OA
total → knee OA
total → fracture
fem. neck → knee OA
knee OA → fracture
knee OA → total
knee OA → fem. neck
0
20
40
60
80
Cochran I² (%)
b
Directional pleiotropy: 1/10
α = 0.05
arms → knee OA
fem. neck → knee OA
legs → knee OA
total → knee OA
total → fracture
knee OA → fracture
knee OA → total
lumbar → knee OA
knee OA → fem. neck
fem. neck → fracture
0
1
2
−log10(MR-Egger intercept P)
c
Outliers flagged
legs → knee OA
fem. neck → fracture
lumbar → knee OA
total → knee OA
fem. neck → knee OA
arms → knee OA
total → fracture
knee OA → fracture
knee OA → total
knee OA → fem. neck
3
3
3
3
3
2
2
2
1
0
0
1
2
3
MR-PRESSO outliers